Mean-field dynamics of Kuramoto oscillators: step-by-step derivation of the Ott-Antonsen reduction

Gonzalo Contreras-Aso

March 12, 2026

1 Preliminaries

Kuramoto model:

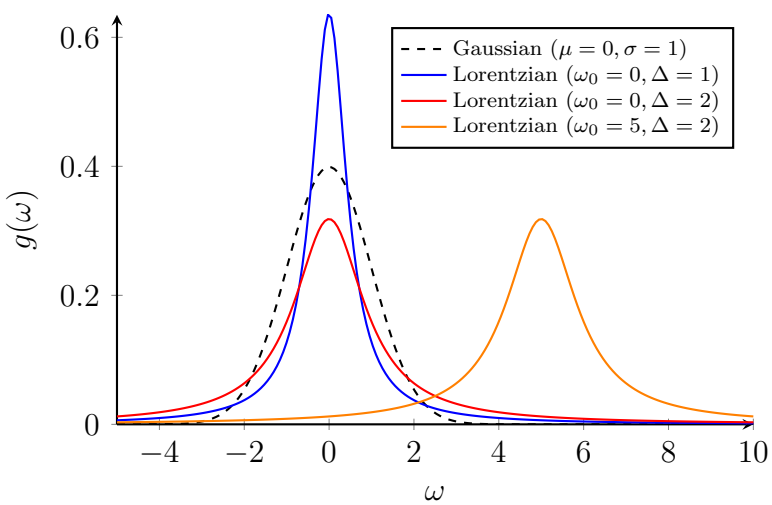
$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i). \tag{1}$$

The intrinsic frequencies ω_i are sampled from a distribution $g(\omega)$. While it is common to use a Gaussian centered at zero, for analytical purposes it is better to consider a Lorentzian distribution (also known as the Cauchy distribution), centered at ω_0:

$$g(\omega) = \frac{\Delta}{\pi}\frac{1}{[(\omega-\omega_0)^2+\Delta^2]} = \frac{1}{2\pi i}\left[\frac{1}{\omega-\omega_0-i\Delta} - \frac{1}{\omega-\omega_0+i\Delta}\right] \tag{2}$$

The parameter Δ is the "half-width at half-maximum", parameterizing how narrow the distribution is around the center, with $\Delta \to 0$ the distribution becomes a Dirac delta function.

Somewhat surprisingly, this distribution has an undefined expectation value and variance.



The advantage of this distribution is the explicit appearance of simple poles, when interpreted as a complex-valued function. The main disadvantage is that this is a fat-tailed distribution, since the extremes of the function do not decay exponentially: if N is large, there will likely be some oscillators with really high/low intrinsic frequencies.

The order parameter r characterizes the mean behavior of the oscillators, with $|r|$ high signaling synchronization:

$$r = \frac{1}{N}\sum_{j=1}^{N} e^{i\theta_j}. \tag{3}$$

In terms of the order parameter we can rewrite the

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + K\,\mathrm{Im}\left(re^{-i\theta_i}\right) = \omega_i + \frac{K}{2i}\left(re^{-i\theta_i} - \bar{r}e^{i\theta_i}\right) \tag{4}$$

Note that this rewriting of the dynamical equation is only valid for the all-to-all coupling (which is the one we consider here since we are interested in mean-field behavior). It is not valid for other topologies.

Note that the upcoming derivation is also applicable [1] (but we will not show it) to

- Coupling to an external frequency Ω (e.g. circadyam rhythms & sunlight)

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i) + \Lambda\sin(\Omega t - \theta_i) \tag{5}$$

- Coupling between communities (σ, σ') of oscillators

$$\frac{\mathrm{d}\theta_i^\sigma}{\mathrm{d}t} = \omega_i + \sum_{\sigma'=1}^{S}\frac{K_{\sigma\sigma'}}{N_{\sigma'}}\sum_{j=1}^{N_{\sigma'}}\sin\left(\theta_j^{\sigma'} - \theta_i^\sigma\right) \tag{6}$$

2 Large N limit and the Ott-Antonsen ansatz

We now consider the "thermodynamic limit", where there are an infinite number of oscillators interacting: In this case, the appropriate description of the system is in terms of the density of oscillators of frequency ω at phase θ at time t, $f(\omega, \theta, t)$. For any time, this function must satisfy

$$\int_0^{2\pi} f(\omega, \theta, t)\mathrm{d}\theta = g(\omega), \tag{7}$$

that is, integrating out the phases we are left with the distribution of possible intrinsic frequencies.

It will be useful to write the Fourier expansion of this density function in the angular variable,

$$f(\omega, \theta, t) = \frac{g(\omega)}{2\pi}\left\{1 + \left[\sum_{n=1}^{\infty} f_n(\omega, t)e^{in\theta} + \text{c.c.}\right]\right\}, \tag{8}$$

where $f_n(\omega, t)$ are the Fourier modes, which clearly satisfies the above condition. In [1] they assumed that these modes can be written as $f_n(\omega, t) \equiv [\alpha(\omega, t)]^n$ (likewise for its complex conjugate), with $|\alpha(\omega, t)| \leq 1$. This is the celebrated *Ott-Antonsen ansatz*.

They motivate this ansatz with two arguments: On the one hand, in the incoherent state $f(\omega, \theta, t) = g(\omega)/2\pi$, and that requires all $f_n(\omega, t)$ to vanish identically. That simply corresponds to $\alpha = 0$ with the ansatz. On the other hand there's stationary synchrony, which also conforms to the ansatz (this is related to a calculation which will come later). Their argument is therefore that the ansatz will connect the two regimes, in which it is clearly valid.

We can write the order parameter in terms of these functions, since it corresponds to the density of oscillators at phase θ at time t, i.e.

$$\begin{aligned} r &= \int_{-\infty}^{\infty} \mathrm{d}\omega \int_0^{2\pi} \mathrm{d}\theta\, f(\omega, \theta, t) e^{i\theta} = \int_{-\infty}^{\infty} \frac{g(\omega)}{2\pi} \mathrm{d}\omega \int_0^{2\pi} \mathrm{d}\theta \left\{ 1 + \left[\sum_{n=1}^{\infty} f_n(\omega, t) e^{in\theta} + \text{c.c.} \right] \right\} e^{i\theta} \\ &= \int_{-\infty}^{\infty} \frac{g(\omega)}{2\pi} \bar{f}_1(\omega, t) 2\pi \mathrm{d}\omega = \int_{-\infty}^{\infty} g(\omega) \bar{\alpha}(\omega, t) \mathrm{d}\omega. \end{aligned} \tag{9}$$

We can already foresee the role of the Lorentzian as in being integrated with the residue theorem alongside a specific contour, which we will discuss later.

2.1 Continuity equation

We now turn to the continuity equation for the density of oscillators: If the density of oscillators at phase θ decreases, that must be due to them having moved (with velocity $\dot{\theta}$) somewhere else. That statement is written as

$$\frac{\partial f}{\partial t} + \frac{\partial}{\partial \theta} \left(\frac{\mathrm{d}\theta}{\mathrm{d}t} f \right) = 0. \tag{10}$$

Inserting the Kuramoto equation (without the indices, since we are working under the continuum approximation), we have

$$\frac{\partial f}{\partial t} - \frac{\partial}{\partial \theta} \left\{ \left[\omega + \frac{K}{2i} \left(r e^{-i\theta} - \bar{r} e^{i\theta} \right) \right] f \right\} = 0. \tag{11}$$

Inserting the Fourier expansion of the density of oscillators (8), along with the Ott-Antonsen ansatz, and grouping by terms of the series, we obtain the following list of equations, which must all be satisfied.

- Zeroth order (e^0):

$$-\frac{K}{2}(r\alpha + \bar{r}\alpha) + \frac{K}{2}(r\alpha + \bar{r}\alpha) = 0. \tag{12}$$

  This is trivially satisfied.

- First orders ($e^{i\theta}, e^{-i\theta}$):
$$\frac{\partial \alpha}{\partial t} + i\omega\alpha + \frac{K}{2}(\alpha^2 r - \bar{r}) = 0, \tag{13}$$
and the conjugate for $\bar{\alpha}$. This constitutes a non-trivial equation for $\alpha(\omega, t)$.

- Second orders ($e^{2i\theta}, e^{-2i\theta}$):
$$2\alpha\left[\frac{\partial \alpha}{\partial t} + i\omega\alpha + \frac{K}{2}(\alpha^2 r - \bar{r})\right] = 0, \tag{14}$$
which (almost magically) yields the exact same equation as before, therefore it does not introduce further constraints. Likewise for the complex conjugate.

- Higher orders: It can be shown that all orders yields the exact same equation, in which the n has been factored out:
$$n\alpha^{n-1}\left[\frac{\partial \alpha}{\partial t} + i\omega\alpha + \frac{K}{2}(\alpha^2 r - \bar{r})\right] = 0. \tag{15}$$
This is what made this work stand out, it collapses the infinite dimensional problem into a single equation.

Therefore, the continuity equation imposes constraints in the sense of dynamical equations for the evolution of the modes, (13) and its complex conjugate.

2.2 Properties of the evolution of the modes

We now for a moment return to the density function (8). Under the Ott-Antonsen ansatz, we can actually sum the geometric series (actually, assuming $|\alpha| < 1$ strictly), and using some trigonometric identities we obtain

$$\begin{aligned} f(\omega, \theta, t) &= \frac{g(\omega)}{2\pi}\left(1 + \sum_{n=1}^{\infty}\left[\alpha(\omega, t)e^{i\theta}\right]^n + \text{c.c.}\right) = \frac{g(\omega)}{2\pi}\left(1 + \frac{\alpha(\omega, t)e^{i\theta}}{1 - \alpha(\omega, t)e^{i\theta}} + \text{c.c.}\right) \\ &= \frac{g(\omega)}{2\pi}\frac{(1 - |\alpha|)(1 + |\alpha|)}{(1 - |\alpha|)^2 + 4|\alpha|\sin^2[(\theta - \Psi)/2]}, \end{aligned} \tag{16}$$

where $\alpha = |\alpha|e^{i\Psi}$. This expression allows us to check several facts:

- The term in the brackets is the Poisson kernel, a classic object in harmonic analysis. Historically, it was known that the stationary synchronous states of the Kuramoto model took this specific functional form, this is something we alluded to when motivating the ansatz. Ott and Antonsen's breakthrough was realizing that if the system starts with a Poisson-kernel distribution, it remains a Poisson kernel for all time, even during the approach to synchrony.

- For $|\alpha| < 1$, the density is positive: $f(\omega, \theta, t) > 0$, as it should.

- For $|\alpha| \to 1$, the density becomes a delta function in the phase variable,

$$f(\omega,\theta,t) \xrightarrow{|\alpha|\to 1} \begin{cases} +\infty & \text{if } \theta = \Psi, \\ 0 & \text{otherwise.} \end{cases} \quad \Rightarrow \quad f(\omega,\theta,t) \to \frac{g(\omega)}{2\pi}\delta(\theta-\Psi) \tag{17}$$

  This represents perfect phase locking.

- The evolution is consistent with $|\alpha| \leq 1$, meaning that if we start in that regime, we remain there. To see this, write the evolution of $\alpha = |\alpha|e^{i\Psi}$ explicitly:

$$\begin{aligned} 0 &= \frac{\partial(|\alpha|e^{i\Psi})}{\partial t} + i\omega|\alpha|e^{i\Psi} + \frac{K}{2}(|\alpha|^2 e^{2i\Psi} r - \bar{r}) \\ &= \frac{\partial|\alpha|}{\partial t}e^{i\Psi} + i|\alpha|\frac{\partial\Psi}{\partial t}e^{i\Psi} + i\omega|\alpha|e^{i\Psi} + \frac{K}{2}(|\alpha|^2 e^{i\Psi} r - \bar{r}e^{-i\Psi})e^{i\Psi}. \end{aligned} \tag{18}$$

  Factor out the exponentials, and take the real part of the expression, remembering that $\mathrm{Re}(re^{i\Psi}) = \mathrm{Re}(\bar{r}e^{-i\Psi})$. One ends up with

$$\frac{\partial|\alpha|}{\partial t} + \frac{K}{2}(|\alpha|^2 - 1)\,\mathrm{Re}(re^{i\Psi}) = 0 \tag{19}$$

  From this expression it is clear that $|\alpha| = 1$ is a fixed point of the dynamics, therefore the trajectory $\alpha(\omega,t)$ is confined to the unit circle.

The next set of assumptions that are required is 1) that $\alpha(\omega,t)$ can be analytically continued from real ω to the complex ω-plane, 2) that this continuation does not have singularities in the lower half plane, 3) that $|\alpha(\omega,t)| \to 0$ as $\mathrm{Im}(\omega) \to -\infty$. These are subtle assumptions but in [1] they argue that, if the initial condition $\alpha(\omega,0)$ satisfies them, they will be satisfied throughout the evolution.

We will not delve into these assumptions here, except for the last one of them, which will be particularly important later: We can see it as a result of equation (13) with $\mathrm{Im}(\omega) \ll 0$, in which case

$$\frac{\partial\alpha}{\partial t} \approx -i\omega\alpha \approx -|\,\mathrm{Im}(\omega)|\alpha \quad \Rightarrow \quad \alpha(\omega,t) \approx \alpha(\omega,0)e^{-|\,\mathrm{Im}(\omega)|t} \xrightarrow{\mathrm{Im}(\omega)\to-\infty} 0 \tag{20}$$

The invariant manifold M on which the dynamics takes place is therefore the space of functions (8) with modes satisfying the three assumptions. It is invariant since, as they showed, if one starts there one remains there. In a later paper [2] they also showed that, under weak conditions, this manifold is also attracting.

2.3 Contour integration

We are now ready to compute the order parameter from the density modes, following the calculation[1] we started in (9). Having analytically extended $\alpha(\omega,t)$ to the complex ω-plane,

[1] Technically, we will show this for its complex conjugate version, since that one involves α and not $\bar{\alpha}$, but analogous arguments apply in that case.

we can compute the relevant integral as part of a contour integral over the lower half-plane, i.e.

$$\oint_{\mathcal{C}} \alpha(\omega,t)g(\omega)\mathrm{d}\omega = \lim_{|\omega|\to\infty}\left(\int_{-|\omega|}^{|\omega|} + \int_{\mathrm{arc}}\right)\alpha(\omega,t)g(\omega)\mathrm{d}\omega \tag{21}$$

Focusing first on the RHS, we see that the first contribution, from the real axis, is the order parameter $\bar{r}$. The second one vanishes: we can parameterize the arc as $\omega(s) = |\omega|e^{-is}$ with $s \in [0,\pi]$, hence

$$I_{\mathrm{arc}} = \lim_{|\omega|\to\infty}\int_0^{\pi}\alpha(\omega,t)g(\omega)\frac{\mathrm{d}\omega}{\mathrm{d}s}\mathrm{d}s = \lim_{|\omega|\to\infty}\int_0^{\pi}\alpha(\omega,t)\frac{\Delta}{\pi}\frac{-i|\omega|e^{-is}}{[(\omega-\omega_0)^2+\Delta^2]}\mathrm{d}s, \tag{22}$$

and, even though there is $|\omega|$ in the numerator, both the denominator and $\alpha(\omega,t)$ go to zero in the limit, hence, $I_{\mathrm{arc}} = 0$.

All that is left for us to do is to compute the contour integral of the LHS: For that, we use the analyticity and lack of singularities of $\alpha(\omega,t)$ in the lower half-plane, which allows us to compute the integral via the residue theorem.

$$\bar{r} = \oint_{\mathcal{C}}\alpha(\omega,t)g(\omega)\mathrm{d}\omega = -2\pi i\sum_{\omega_p}\mathrm{Res}_{\omega=\omega_p}\{\alpha(\omega,t)g(\omega)\}, \tag{23}$$

where the minus sign is due to the contour being chosen as clockwise.

The simple poles ω_p come from the Lorentzian (2); more concretely the one within the contour is $\omega_0 - i\Delta$, see Figure 1 for a visualization of the contour and the poles.

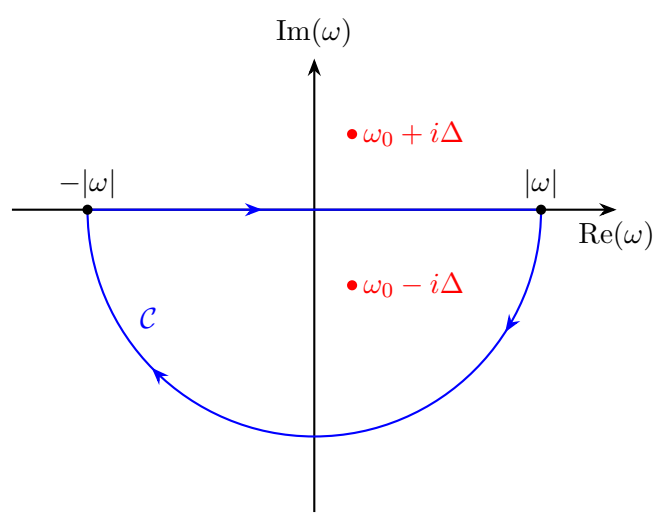


Figure 1: Visualization of the contour $\mathcal{C}$ and the simple poles of $g(\omega)$.

We thus arrive at the relation $\bar{r} = \alpha(\omega_0 - i\Delta, t)$. We now perform, without loss of generality a change of variables $(\theta,\omega) \to (\theta - \omega_0 t, (\omega-\omega_0)/\Delta)$. The phase change is essentially a change to the rotating frame, while the frequency part leaves $g(\omega)\mathrm{d}\omega \to g(\omega')\mathrm{d}\omega'$ invariant.

That change is equivalent to setting $\omega_0 = 0$, $\Delta = 1$, and thus $\bar{r} = \alpha(-i, t)$, and similarly $r = \bar{\alpha}(-i, t)$.

Back to the equation for the evolution of the modes (13), we can relate it to the equation for the order parameter using the relation we just found, setting $\omega \to -i$.

$$\frac{\partial \bar{r}}{\partial t} + \bar{r} + \frac{K}{2}\bar{r}(|r|^2 - 1) = 0. \tag{24}$$

3 The order parameter equation

Finally, if we write the order parameter in polar form $r = \rho(t)e^{i\Psi(t)}$, we arrive at the following set of ordinary differential equations,

$$\frac{\partial \rho}{\partial t} + \left(1 - \frac{K}{2}\right)\rho + \frac{K}{2}\rho^3 = 0, \quad \frac{\partial \Psi}{\partial t} = 0. \tag{25}$$

This not only reduced the infinite-dimensional problem (with an equation per mode in the Fourier expansion of the density function) to a lower dimensional problem consisting of two uncoupled ordinary differential equations, but it also has an analytical solution:

$$\frac{\rho(t)}{R} = \left|1 + \left[\left(\frac{R}{\rho(0)}\right)^2 - 1\right]\exp[(1 - K/2)t]\right|^{-1/2}, \quad R = \sqrt{|1 - 2/K|}. \tag{26}$$

There are some notable properties of this solution:

- The critical coupling is identified as $K_c = 2$.
- For $K < K_c$, the order parameter goes to zero, $\rho(t) \rightsquigarrow 1/(1 + \exp(+\infty)) = 0$.
- For $K > K_c$, the order parameter asymptotes to the finite value $\sqrt{1 - 2/K}$.

This is all in accordance with the previously known results that $K_c = 2/(\pi g(0))$ for the Kuramoto model with Lorentzian frequencies, with the two phases described above [3].

References

[1] Edward Ott and Thomas M Antonsen. "Low dimensional behavior of large systems of globally coupled oscillators". In: *Chaos: An Interdisciplinary Journal of Nonlinear Science* 18.3 (2008).

[2] Edward Ott and Thomas M Antonsen. "Long time evolution of phase oscillator systems". In: *Chaos: An interdisciplinary journal of nonlinear science* 19.2 (2009).

[3] Steven H Strogatz. "From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators". In: *Physica D: Nonlinear Phenomena* 143.1-4 (2000), pp. 1–20.